Project Navi

Bachelor Designer & Developer

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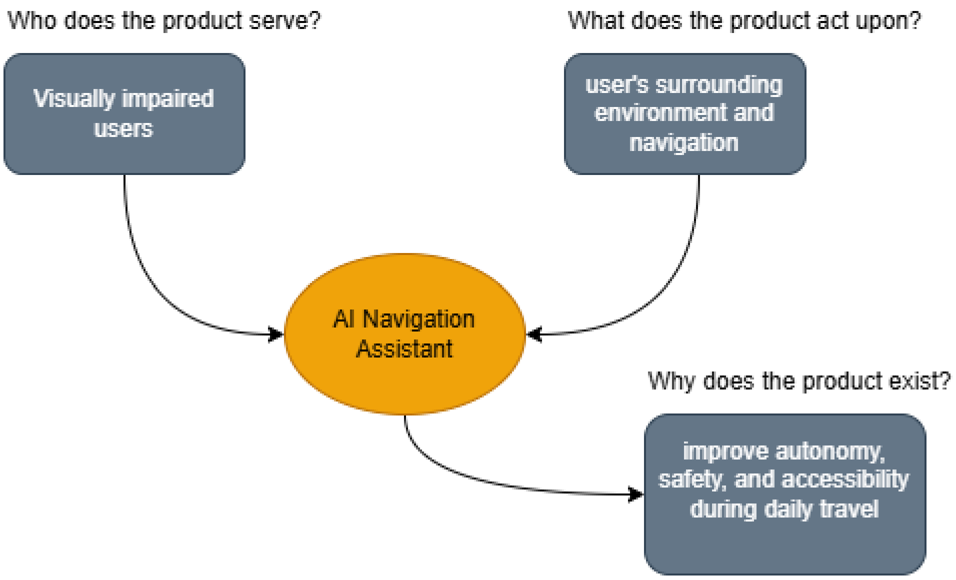
Functional Analysis

Needs Analysis

The purpose of this project is to develop a mobile application that assists visually impaired users in their daily navigation. The application aims to improve user autonomy by providing real-time voice guidance, obstacle detection, and environmental descriptions through artificial intelligence and GPS technologies.

Horned Beast Diagram



The Horned Beast Diagram, which summarizes the fundamental need addressed by the project. The application is intended for visually impaired users and acts upon their surrounding environment by providing navigation assistance and obstacle awareness. Its primary objective is to improve user autonomy, safety, and accessibility during daily travel.

Validation of the Need

Before defining the application's functionalities, it is important to verify that the identified need is relevant and sustainable. This validation consists of analyzing the arguments supporting the need for the application as well as the potential limitations or counterarguments. If at least one supporting argument remains valid, the need can be considered justified.

Need Validation Table

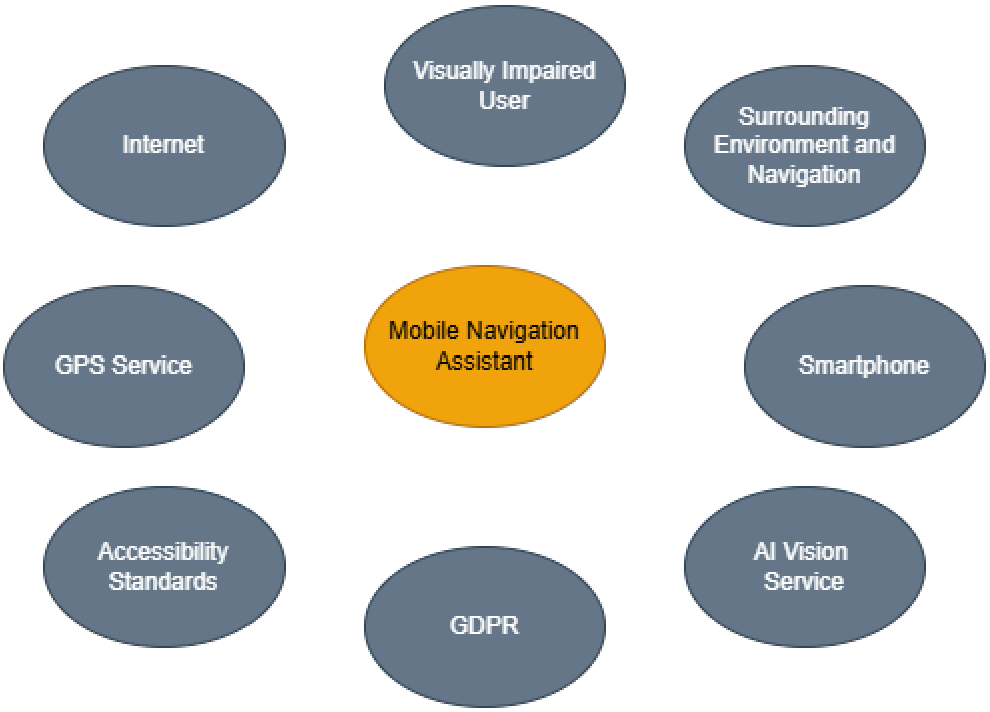
Supporting Arguments	Counterarguments
Visually impaired people need greater autonomy in their daily mobility.	Existing navigation applications already provide guidance.
Current applications are not specifically designed for visually impaired users.	Some assistive technologies are already available.
Artificial intelligence can improve obstacle detection and environmental awareness.	AI systems may occasionally produce inaccurate results.
Smartphones are widely available and include GPS, cameras, and voice assistants.	The application depends on the smartphone's battery and internet connection.
Improving accessibility contributes to greater social inclusion and independence.	Some users may still require human assistance in complex situations.

Although alternative navigation solutions already exist, they do not fully address the specific needs of visually impaired users. The combination of artificial intelligence, voice interaction, and real-time navigation offers significant added value. Therefore, the identified need is considered valid and sustainable.

Product Environment Analysis

The product environment analysis identifies all external elements that interact with the application during its operation. These interactors represent the users, systems, devices, regulations, and services that influence the application's functionality and performance. Understanding these interactions helps define the service and constraint functions in the next stage of the functional analysis.

Interactor Map

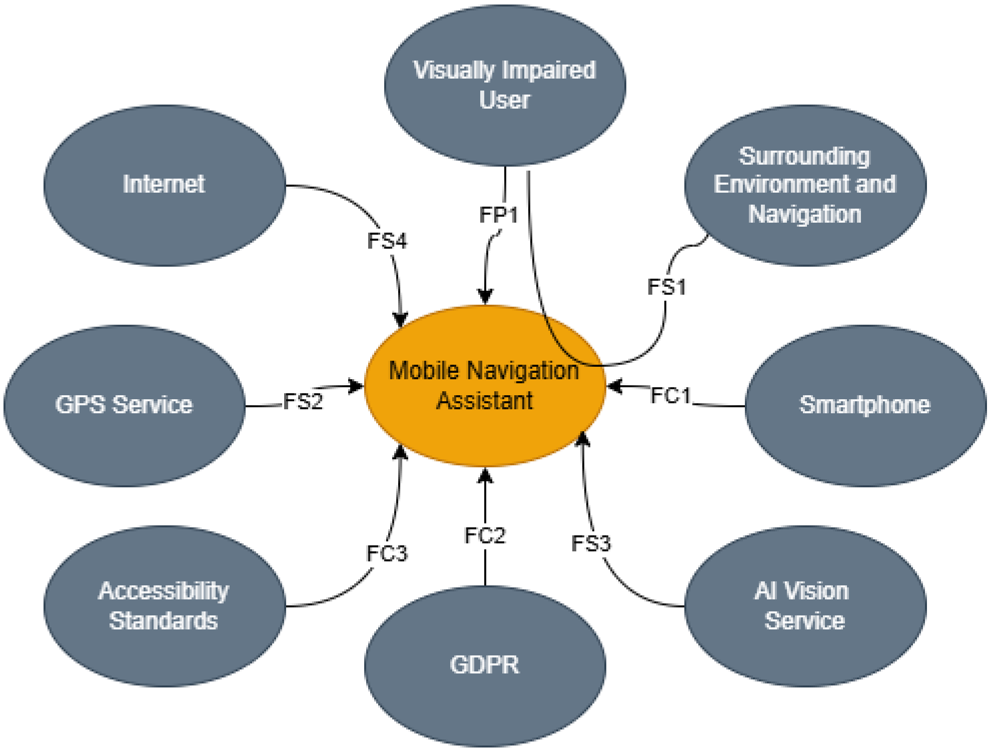


The interactor map illustrates the different entities interacting with the application. It highlights the relationships between the visually impaired user, the smartphone, external services such as GPS and artificial intelligence, as well as legal and technical constraints. These interactions provide the foundation for identifying the application's service and constraint functions.

Functional Analysis

The functional analysis identifies the services that the application must provide and the constraints it must satisfy in relation to its environment. These functions are represented using an Octopus Diagram, which illustrates the interactions between the product and each identified interactor.

Octopus Diagram



FP1	Assist visually impaired users during navigation
FS1	Guide the user through voice assistance
FS2	Determine the user's location
FS3	Analyze and describe the surroundings
FS4	Access online navigation services
FC1	Operate on a compatible mobile device
FC2	Ensure personal data protection
FC3	Comply with accessibility requirements

The Octopus Diagram, which illustrates the interactions between the Mobile Navigation Assistant and its environment. The diagram distinguishes **service functions**, which describe the services provided to the user and external systems, from **constraint functions**, which represent the technical, legal, and operational requirements that the application must satisfy.

Functional Characterization

Ref.	Function	Value	Criteria		Target Level	Flexibility	Limits
FP1	Assist visually impaired users during navigation	4	FP1c1	Navigation guidance accuracy	Safe and reliable navigation	F0	Depends on GPS and AI accuracy
			FP1c2	Voice guidance quality	Clear real-time instructions	F1	Speaker or headphones required
			FP1c3	User safety	No critical navigation errors	F0	Limited by unexpected obstacles
FS1	Guide the user through voice assistance	4	FS1c1	Voice clarity	Easily understandable speech	F1	Ambient noise may reduce intelligibility
			FS1c2	Response time	Less than 1 second	F1	Depends on smartphone performance
FS2	Determine the user's location	4	FS2c1	GPS accuracy	≤ 5 meters	F0	Indoor environments may reduce accuracy
			FS2c2	Position update frequency	Continuous updates	F1	GPS signal availability
FS3	Analyze and describe the surroundings	4	FS3c1	Obstacle detection	Real-time detection	F1	Camera quality and lighting conditions
			FS3c2	Object recognition	≥ 90% recognition accuracy	F2	Weather and image quality
FS4	Access online navigation services	3	FS4c1	Internet availability	Stable connection	F2	Limited functionality in offline mode
			FS4c2	Map synchronization	Automatic updates	F2	Depends on network availability
FC1	Operate on a compatible mobile device	3	FC1c1	Device compatibility	Android and iOS supported	F1	Requires supported OS versions
FC2	Ensure personal data protection	4	FC2c1	GDPR compliance	100% compliant	F0	Legal requirements must always be respected
			FC2c2	Data encryption	End-to-end encryption	F0	Secure storage required
FC3	Comply with accessibility requirements	4	FC3c1	WCAG compliance	WCAG 2.2 AA	F0	Must remain compliant after updates
			FC3c2	Screen reader compatibility	Fully compatible	F0	Depends on the operating system

The functional characterization defines the expected performance of each identified function. For every primary, service, and constraint function, evaluation criteria, target levels, flexibility, and limits are specified to ensure that the Mobile Navigation Assistant meets the needs of visually impaired users while respecting technical and regulatory requirements.

Function Table

Functional Analysis Summary

The functional analysis allowed the identification and characterization of the main functions required for the Mobile Navigation Assistant. The study highlighted the primary objective of the application: assisting visually impaired users during outdoor navigation while ensuring safety, accessibility, and reliability.

The Octopus Diagram identified the interactions between the product and its environment, while the functional characterization table defined the expected performance levels and constraints associated with each function.

These functional requirements will serve as a foundation for the next stages of the project, particularly the technical design phase and the creation of the detailed specifications.